INCH-POUND

A-A-54795
11 January 1993

COMMERCIAL ITEM DESCRIPTION

WATER TESTING KIT, BACTERIOLOGICAL

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-W-36258D which is cancelled.

This Commercial Item Description covers a bacteriological water analysis kit, suitable for determining the potability of drinking water.

Salient characteristics:

General. Shall be a self-contained bacteriological water analysis kit using the membrane filter technique for determining total coliform in water under field conditions. The water analysis kit shall include a sanitarian's kit, and an incubator that is electrically operated and includes multi-voltage selection. The sanitarian's kit shall be designed to vacuum filter water samples through microporous filters. The incubator shall be operable from multi-voltage supply lines that include 12 and 24 volts DC, and 115 and 230 volts, 50/60 Hz AC. The incubator shall be operable in ambient temperatures ranging form -30°F to 95°F. The incubator shall have the capacity to simultaneously accommodate 24 test samples that are held in 12 x 50 mm commercial size plastic Petri culture dishes that include media absorbent pads.

Design and construction. The design and construction of the water analysis kit shall conform to Figures 1 and 2.

The complete water analysis kit shall be enclosed in a fiberglass-reinforced plastic carrying case (container) of such size and weight as to permit hand carrying by one man. The kit shall be provided with an incubator with operating controls, an incubator rack, a filter holder assembly with supporting base, a funnel assembly, a filter support assembly, a vacuum filter flask, repair parts and tools.

AMSC N/A FSC 6665

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Carrying case. The carrying case shall have maximum overall dimensions (without handle) of 15 inches long by 10 inches wide by 10 inches high. All corners and edges shall be round. Case shall be moisture and mold-resistant throughout, and shall be capable of withstanding rough handling under field conditions. A plastic, collapsible carrying handle shall be securely mounted on top of the cover.

The carrying case shall be provided with a sufficient number of pull/down corrosion-resisting metal clamps of suitable size and tension to afford a rainproof and slashproof closure. The carrying case shall be so designed that the cover may be opened to a maximum position of 90° without the front end of the case lifting from the surface.

A plastic holding device (insert) shall be mounted on the inside of the cover to hold the following components securely in place:

A Sanitarian's syringe for cutout "A".

A 4 ounce polyethylene screw cap bottle for cutout "B".

A graduated sampling cup for cutout "C".

A package of Endo broth for cutout "D".

An 8 ounce polypropylene bottle with phenolformaldehyde screw cap for cutout "E".

As indicated in Figure 1, a pressure-sensitive plate shall be securely mounted on the outside of the carrying case at a position alongside the receptacle. The plate shall be permanently and legibly marked in red with the following or a similar legend:

"CAUTION: Set voltage selector switch prior to plugging in power cord."

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Incubator controls. The incubator controls shall be capable of maintaining an incubation temperature of 35°C ± 0.5°C within the incubator cavity in ambient temperatures varying from -30°F to 95°F. A voltage selector switch shall allow the incubator to operate from 12 and 24 volt DC supply lines, and from 115 and 230 volt 50/60 Hz AC supply lines. As indicated in Figure 1, a control panel shall be mounted in the carrying case to hold the voltage selector switch, a temperature control dial, a pilot light, and a fuse holder. A male "twist-lock" type receptacle shall be mounted on the outside of the carrying case, to accommodate an external line cord for the voltage supply. The receptacle shall be gasket-mounted and recessed, and shall be supplied with a captive cover for protective covering when not in use. A bellows type thermostat shall be included for the incubator temperature control. The bulb of the thermostat shall include a protective metal shield and shall be located in the bottom of the incubator cavity. The front and rear of the control panel shall include recesses to house the power line cord and container for holding accessories and repair parts.

Incubator rack. A square corrosion-resisting wire rack conforming to Figure 2 shall be provided for holding Petri culture dishes intended for simultaneous incubation. The incubator rack shall have the overall capacity to accommodate 24 12 x 50 mm commercial size Petri culture dishes. The cross wire construction shall form wells to accommodate not less than 48 13 x 100 mm size test tubes in an upright position with the bottom shelf as a support for the test tube bottoms. The frame, as a minimum dimension, shall be formed from 0.134 inch diameter (No. 10 Birmingham (B.W.) gauge) wire. The cross wires, as a minimum dimension, shall be formed from 0.083 inch diameter (No. 14 B.W. gauge) wire. The wires shall be securely welded or brazed in position.

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Sanitarian's kit. A sanitarian's kit shall be supplied, and shall consist of a metal syringe with a two-way valve, and a sampling cup. The syringe and sampling cup shall conform to Figure 2 and shall be dimensioned to fit into cutouts "A" and "C" respectively of the carrying case cover insert of Figure 1. The syringe with plunger in the closed position shall be 7-1/2 inches long and shall be fabricated from chrome-plated brass or corrosion-resisting steel. The barrel of the syringe shall include legible instructions for using the syringe. A 7-1/2 inch long fluorelastomer tube shall be attached to the inlet of the syringe by means of a luer adapter type termination. The other end of the fluorelastomer shall terminate with a nylon luer-tapered adapter. The sampling cup shall be fabricated from corrosion-resisting steel that is .0007 to .017 inch thick, and shall include graduations indicated in Figure 2. The external surface of the sampling cup shall have a No. 7 commercial mirror finish. The inside surface shall have a satin finish. The sampling cup shall be used to collect a water test sample, and shall have a capacity of 250 ml.

Filter holder support assembly. A filter holder support assembly shall be supplied to vacuum filter the water sample through an inserted microporous disc filter. The attachment of the valved syringe (from the sanitarian's kit) shall provide the vacuum pump action. The filter holder support assembly shall consist of the following:

A funnel assembly with a locking ring (see Figure 1.

A filter holder assembly with a support base and a removable screen (see Figure 1).

A beaker type vacuum filter flask (see Figure 1).

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The filter holder support assembly shall be designed to be assembled for filtering water samples, in accordance with the following:

A microporous filter is placed on the screen of the support base with the grid side of the filter in the "up" position.

The funnel assembly is attached to the filter holder assembly and locked in place by turning its locking ring to form a leak proof seal.

The supporting base of the filter holder assembly is fitted into the vacuum filter flask to form a leakproof seal.

The support base of the filter holder assembly shall be sterilizable by the action of formaldehyde bactericidal gas produced by an incomplete combustion of methyl alcohol. The formaldehyde bactericidal gas shall be producible in accordance with the following procedure:

The funnel assembly without a microporous filter disc is attached to the filter holder assembly.

The wick attached to the filter holder is moistened with methyl alcohol and then ignited.

The vacuum filter flask is placed face down over the burning methyl alcohol to produce the bactericidal formaldehyde gas.

Funnel assembly. The funnel assembly shall consist of a corrosion-resisting steel 100 ml capacity (minimum) funnel, 0.036 inch minimum thick of a general funnel design. The base shall be collared with an aluminum locking ring with nylon plastic rollers (wheels) suitably positioned on the inside. These rollers shall facilitate a twist-type locking action of the locking ring. The filter support assembly shall be so designed that turning the locking ring clockwise shall cause the funnel assembly to lock securely in place and afford a leakproof seal. A vacuum of 25-26 inches of mercury in the system shall not drop below 20 inches of mercury within a period of 20 minutes, when tested, as specified in the Quality Assurance Provisions for "Leakage test".

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Filter holder assembly. The filter support assembly shall be fabricated of smooth finish corrosion-resisting steel and shall consist of a removable screen (100-mesh minimum) assembly and the filter base upon which it shall be centrally positioned as shown in Figure 2.

Supporting base. The supporting base shall be fabricated of aluminum alloy of sufficient weight to afford stability for the complete filter holder assembly, and shall be in accordance with Figure 2. A wick composed of a woven non-asbestos fiber and is 1/4-inch nominal width shall be contained in a recess around the top of the base and shall serve to sterilize the filtering components by means of formaldehyde gas (produced by the incomplete combustion of methyl alcohol). Two Buna-N-Rubber O-rings shall be contained in groves around the perimeter of the base to afford a complete seal with the vacuum filter flask. An outlet evacuator hole of luer taper design shall connect to the bottom inner well of the supporting base. Three or four nylon plastic feet shall be provided in a manner to afford stability to the supporting base. There shall be two release grooves on the perimeter to provide assistance in removing the support base from the vacuum filter flask.

Vacuum filter flask. The vacuum filter flask shall be a corrosion-resisting steel beaker, 0.031 inch thick (No. 22 U.S. Standard gauge) minimum, and shall be not less than 590 ml capacity. The rim of the filter flask shall be so formed that it permits formaldehyde gas sterilization and also forms a leakproof seal with the two O-rings on the sides of the supporting base.

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Accessories, repair parts and tools. The following shall be furnished with the water testing kit:

One (1) each Forceps, Straight, Flattened Tips, 4-1/2 inches, corrosion-resisting steel.

One (1) each Pencil, Marking, Red for glazed surface marking.

One (1) Cord, Electrical, 3-conductor, No. 18AWG, Type SVT or SV with a 3-conductor male plug at one end and a twist lock female receptacle at the other end for mating with the recessed receptacle on the case.

One(1) each Container, Plastic, translucent, flexible, containing the following:

One (1) each Lamp, indicator, clear, 14.4 volts, G 3-1/2 53, bayonet type.

Five (5) each Fuse, cartridge, AGC, 5 amperes, 250 volts.

One (1) each Adapter plug, continental.

Wrench, Allen for control knobs.

Battery adapter, alligator clip assembly.

Envelope containing two (2) nylon lockwheels, two (2) No. 4-40 thread by 1/4 inch long flathead screws, two (2) Allen set screws, and one (1) Allen wrench for set screws.

Workmanship. The water testing kit shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One water testing kit, as specified, constitutes one unit.

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QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspections and testing. Inspections and testing shall be conducted on the water testing kit in accordance with commercial practice. Sampling for the inspections and testing shall be conducted in accordance with commercial practice. Tests shall also be included to determine compliance with specification requirements that include, but shall not be limited to, the following:

Operation Test. Each water testing kit shall be electrically checked to demonstrate its proper operation at each voltage level on the voltage selector switch.

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Leakage test. A filter holder support assembly shall be selected and assembled by securing the supporting base of the filter holder assembly into the vacuum filter flask to form a leakproof seal. The outlet evacuator hole on the supporting base shall be connected by a tube to a vacuum pump with a capability of producing a vacuum of 25 inches of mercury. A 0.45 micron pore size filter shall be completely saturated with water and then placed into the filter holder. The vacuum pump connected to the supporting base shall be operated to obtain a vacuum of 25-26 inches of mercury inside the filter holder assembly. After obtaining the 25-26 inches of mercury vacuumatic pressure, the connecting tube shall be clamped to prevent any flow of air through the evacuator hole, and the vacuum pump operation shall then be stopped. After a 20 minute waiting period, the vacuum inside the filter holder shall be measured by a suitable instrument. A filter holder whose vacuumatic pressure has dropped below 20 inches of mercury within the 20 minute period shall be cause for rejection. Sampling for the testing shall be conducted in accordance with S-1 of MIL-STD-105. The unit of product shall be a filter holder support assembly and the acceptance number shall be zero.

Incubator temperature test. The water testing kit shall be assembled with the incubator rack holding 24 12 x 50 mm plastic Petri dishes that are evenly spaced within the three shelves. Each Petri dish shall contain an absorbant pad, and either a nutrient saturated filter disk or 2 ml of distilled water. At least three Petri dishes on each shelf shall also contain an inserted thermocouple for measuring the surface temperature of a nutrient saturated filter or the water soaked absorbant pad. The water testing kit shall be electrically operated in a conditioned chamber maintained at -29° to -31°F, in accordance with the following:

An external voltage is applied through the voltage selector switch.

The incubator temperature is adjusted in accordance with manufacturer's instructions for a working temperature of 35° to 37°C.

The thermocouple monitored temperatures shall be measured and recorded in 1/2 hour intervals for at least 4 hours. The incubator temperature test shall be repeated with the water testing kit in a conditioned chamber maintained at 93° to 95°F.

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A water testing kit that does not maintain its operating temperature at 35° to 37°C shall be cause for rejection. Sampling for testing shall be in accordance with S-1 of MIL-STD-105. The acceptance number shall be zero.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

NSN	Item Identification
6665-00-682-4765	WATER TESTING KIT.

The following National Stock Numbers cover consumables suitable for use with the water testing kit covered by this document:

NSN	Item Identification
6505-00-687-8436	ENDO BROTH MEMBRANE FILTER, 2 cc, 24s.
6640-00-299-8692	DISK, FILTERING, MICROPOROUS, Hydrosol Type, 47 mm Diameter, 100s.
6640-01-030-9012	DISH, CULTURE, PETRI, Top and Bottom, Disposable, 500s.

MILITARY INTERESTS:

Custodians:

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Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

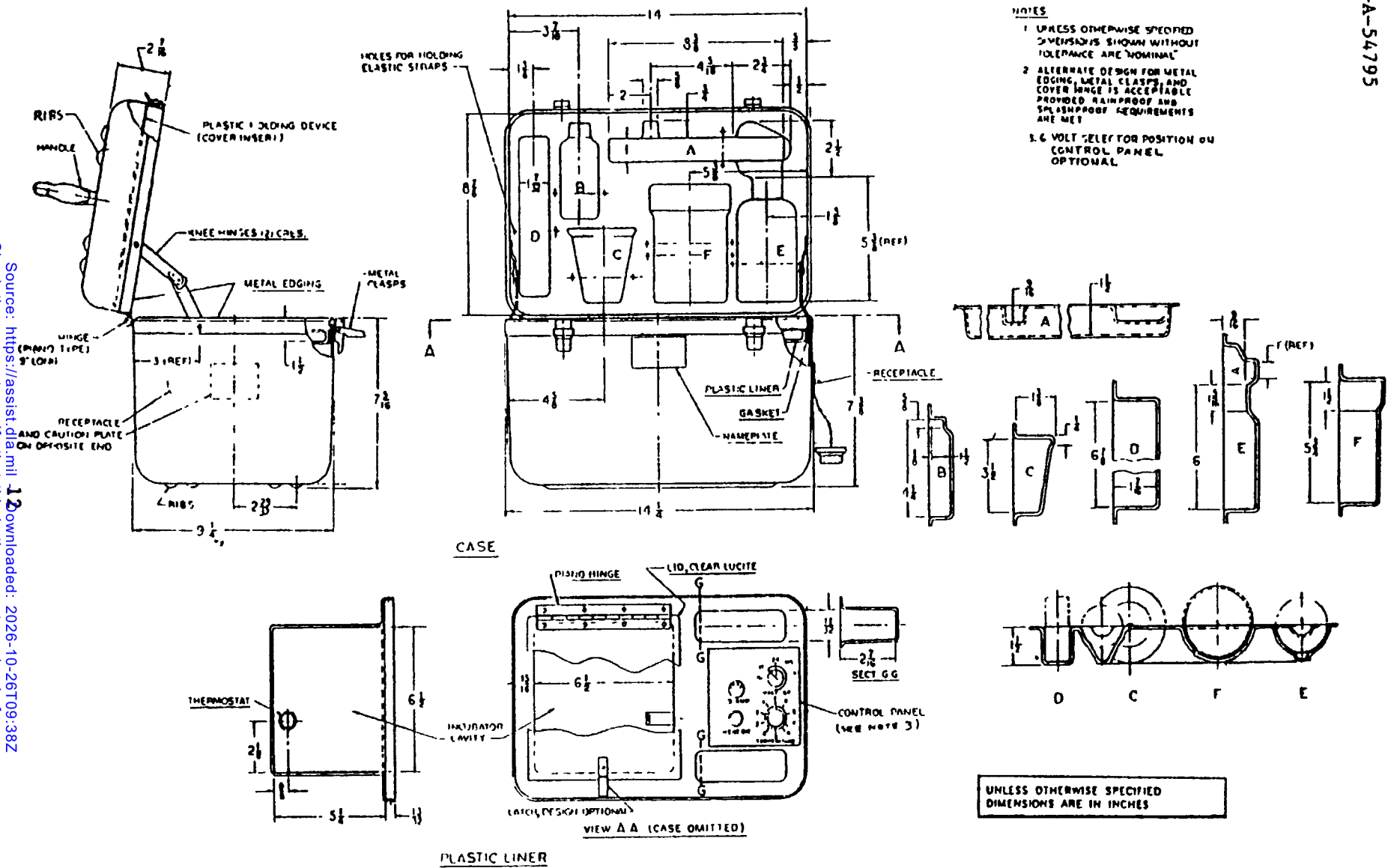
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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
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FDA-MPQAS

Project No. 6665-0635

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NSN 6665-00-682-4765

FIGURE 1 WATER TESTING KIT, BACTERIOLOGICAL

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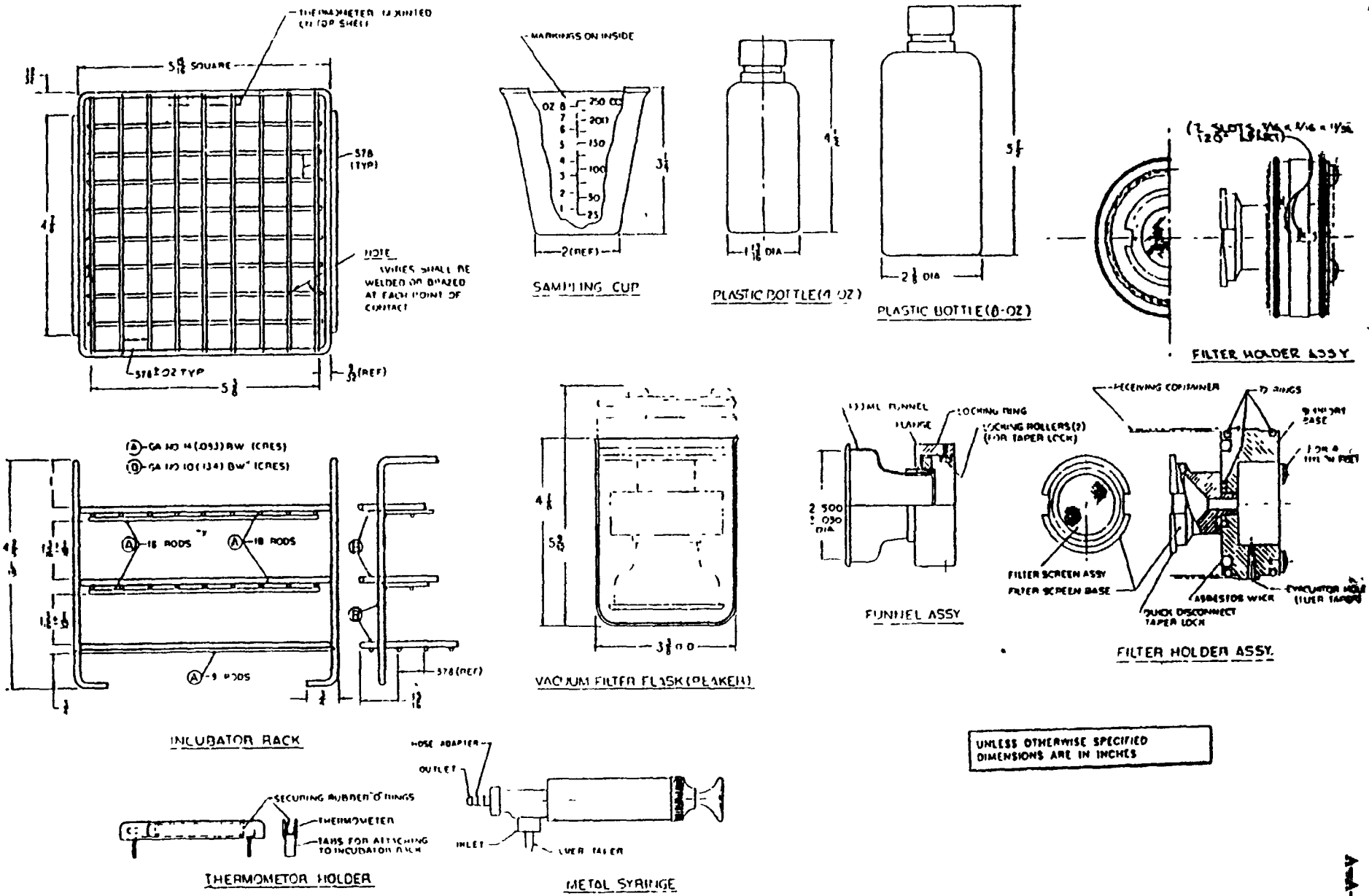


FIGURE 2 WATER TESTING KIT, BACTERIOLOGICAL

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER A-A-54795	2 DOCUMENT DATE (YYMMDD) 11 JANUARY 1993

3. DOCUMENT TITLE

WATER TESTING KIT, BACTERIOLOGICAL

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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